

CLOUD COMPUTING

NAME – ANKUR SHARMA

SRN – PES2UG23CS077

Ss1:

```
C:\Users\ankur\Desktop\SEM6\cc-lab8\PES2UG23CS077\raft_docker_lab>dir
Volume in drive C is OS
Volume Serial Number is DA7C-34F6

Directory of C:\Users\ankur\Desktop\SEM6\cc-lab8\PES2UG23CS077\raft_docker_lab

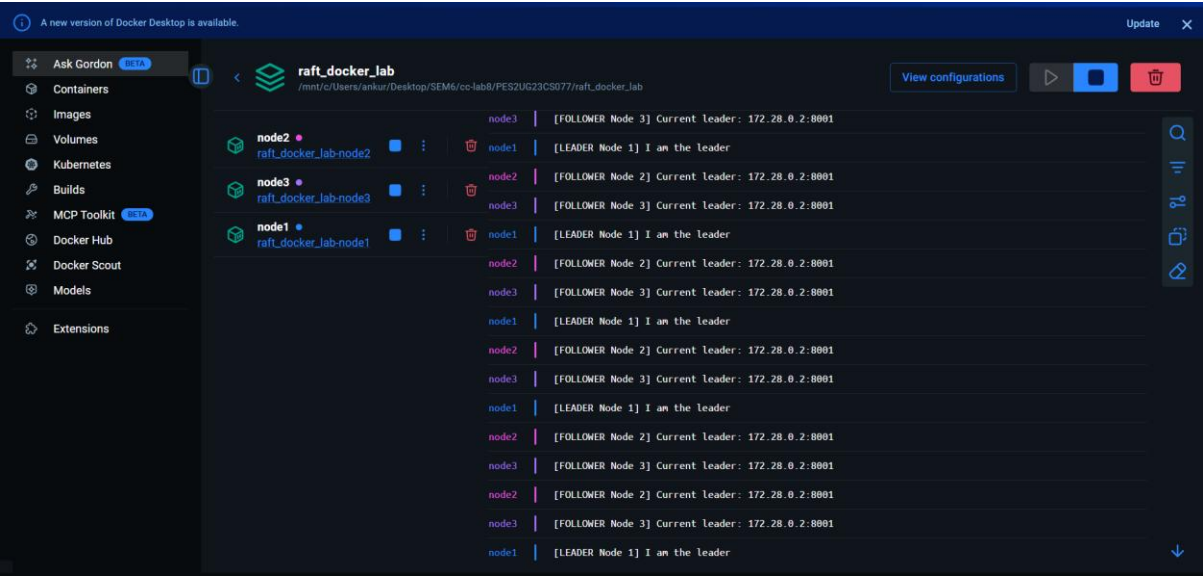
26-03-2026  02.12 PM    <DIR>          .
26-03-2026  02.14 PM    <DIR>          ..
26-03-2026  02.12 PM                562 docker-compose.yml
26-03-2026  02.12 PM                450 Dockerfile
26-03-2026  02.12 PM            1,377 node.py
26-03-2026  02.12 PM                6 requirements.txt
                4 File(s)            2,395 bytes
                2 Dir(s)  7,989,678,080 bytes free

C:\Users\ankur\Desktop\SEM6\cc-lab8\PES2UG23CS077\raft_docker_lab>wsl
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
```

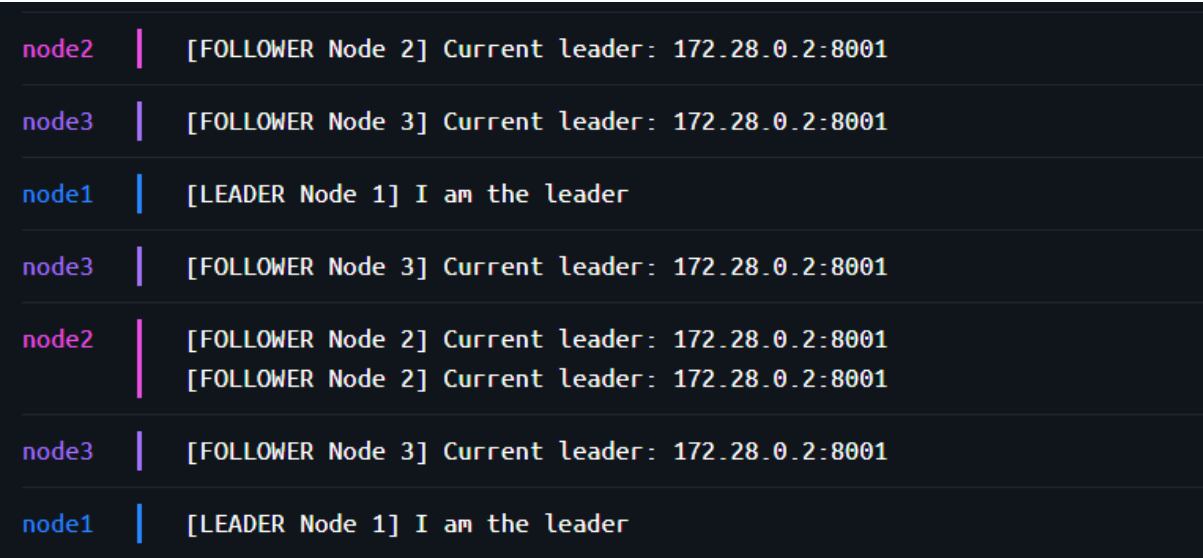
SS2A:

```
-- [nodes] resolving provenance for metadata file
[+] up 7/7
✓Image raft_docker_lab-node3      Built      12
✓Image raft_docker_lab-node1      Built      12
✓Image raft_docker_lab-node2      Built      12
✓Network raft_docker_lab-raftnet  Created    0.
✓Container node3                  Created    0.
✓Container node2                  Created    0.
✓Container node1                  Created    0.
Attaching to node1, node2, node3
node3 | Starting Node 3 at 172.28.0.4:8003
node3 | [Node 3] Waiting for leader election...
node2 | Starting Node 2 at 172.28.0.3:8002
node2 | [Node 2] Waiting for leader election...
node1 | Starting Node 1 at 172.28.0.2:8001
node1 | [Node 1] Waiting for leader election...
node3 | [Node 3] Leader changed: 172.28.0.2:8001
node3 | [FOLLOWER Node 3] Current leader: 172.28.0.2:8001
node2 | [Node 2] Leader changed: 172.28.0.2:8001
node2 | [FOLLOWER Node 2] Current leader: 172.28.0.2:8001
node1 | [Node 1] Leader changed: 172.28.0.2:8001
node1 | [LEADER Node 1] I am the leader
node3 | [FOLLOWER Node 3] Current leader: 172.28.0.2:8001
node2 | [FOLLOWER Node 2] Current leader: 172.28.0.2:8001
```

SS2B:



SS3A :



SS3B :

[illegible]

SS3C:

[illegible]

SS3D:

[illegible]

SS4 :

```

ankur@JITU: /mnt/c/Users/ankur/Desktop/SEM6/cc-lab8/PES2UG23CS077/raft_docker_lab$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS          NAMES
f95bd992b827   raft_docker_lab-node2              "python node.py 2"     13 minutes ago Up 12 minutes          node2
a86b7e0bc051   raft_docker_lab-node3              "python node.py 3"     13 minutes ago Up 12 minutes          node3
b949338b95dc   raft_docker_lab-node1              "python node.py 1"     13 minutes ago Up 12 minutes          node1
ankur@JITU: /mnt/c/Users/ankur/Desktop/SEM6/cc-lab8/PES2UG23CS077/raft_docker_lab$

```

SS5:

```
ankur@JITU: /mnt/c/Users/ankur/Desktop/SEM6/cc-lab8/PES2UG23CS077/raft_docker_lab$ docker network inspect raft_docker_lab
[
  {
    "Name": "raft_docker_lab_raftnet",
    "Id": "e52bc0a17fb23826fc7feba7b02a327ffe4161364af7f3e723098b534451c845",
    "Created": "2026-03-26T08:54:16.229732753Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.28.0.0/16",
          "Gateway": "172.28.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false
  }
]
```

```
ankur@JITU: /mnt/c/Users/an  ×  +  ▾

    "Options": {
      "com.docker.network.enable_ipv4": "true",
      "com.docker.network.enable_ipv6": "false"
    },
    "Labels": {
      "com.docker.compose.config-hash": "7b955128463677559d4edba13e9fe8f69dd803b2dfc190dd0f2333946f",
      "com.docker.compose.network": "raftnet",
      "com.docker.compose.project": "raft_docker_lab",
      "com.docker.compose.version": "5.0.2"
    },
    "Containers": {
      "a86b7e0bc051f1a041543ff0703ad51a182438d25293c466af3cd5c50daa0149": {
        "Name": "node3",
        "EndpointID": "41f09c127bb193a272a9ea06b07c1bbc17d1dac989c52ed74ed47caeab0fbe02",
        "MacAddress": "aa:71:8f:90:13:30",
        "IPv4Address": "172.28.0.4/16",
        "IPv6Address": ""
      },
      "b949338b95dc211d9fe4519c86a0e0a92531c9b5f485b7008480687371105f41": {
        "Name": "node1",
        "EndpointID": "ada2025730557ad4a96409342264e16209220fbf44ceee0f43cf12ab71db1e32",
        "MacAddress": "76:cc:ca:2b:50:79",
        "IPv4Address": "172.28.0.2/16",
        "IPv6Address": ""
      },
      "f95bd992b827b20868fe5faf7f96734d76b4ac2897a9ddd0803450117c140dba": {
        "Name": "node2",
        "EndpointID": "17d49112f6b4cd38909877e73fa91d1f5f795d1f989e782f11fb946d3d42339b",
        "MacAddress": "f2:a1:b1:d7:95:20",
        "IPv4Address": "172.28.0.3/16",
        "IPv6Address": ""
      }
    }
  }
]
```


SS6C:

```
[FOLLOWER Node 3] Current leader: 172.28.0.2:8001
[Node 3] Leader changed: None
[Node 3] Waiting for leader election...
[Node 3] Leader changed: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
```

SS7A:

node1	[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
node3	[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
node2	[LEADER Node 2] I am the leader
node1	[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
node2	[LEADER Node 2] I am the leader
node3	[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
node1	[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
node2	[LEADER Node 2] I am the leader
node3	[FOLLOWER Node 3] Current leader: 172.28.0.3:8002

SS7B:

```
Starting Node 1 at 172.28.0.2:8001
[Node 1] Waiting for leader election...
[Node 1] Leader changed: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
```

ACTIVITY :

SS8:

```
ankur@JITU: /mnt/c/Users/ankur/Desktop/SEM6/cc-lab8/PES2UG23CS077/raft_docker_lab$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED        STATUS        PORTS          NAMES
9a1c1d61ef5e   raft_docker_lab-node1 "python node.py 1"      58 seconds ago Up 57 seconds          node1
c6385eb8b544   raft_docker_lab-node3 "python node.py 3"      58 seconds ago Up 57 seconds          node3
40e1cbad0432   raft_docker_lab-node4 "python node.py 4"      58 seconds ago Up 57 seconds          node4
0c81f90ae207   raft_docker_lab-node5 "python node.py 5"      58 seconds ago Up 57 seconds          node5
9bd2dc5543     raft_docker_lab-node2 "python node.py 2"      58 seconds ago Up 57 seconds          node2
ankur@JITU: /mnt/c/Users/ankur/Desktop/SEM6/cc-lab8/PES2UG23CS077/raft_docker_lab$
```


SS9:

[illegible]

SS10:

[illegible]

SS11:

[illegible]

SS12:

[illegible]

SS13:

[illegible]

SS14:

```
ankur@JITU: /mnt/c/Users/ankur/Desktop/SEM6/cc-lab8/PES2UG23CS077/raft_docker_lab$ docker stop node4
node4
ankur@JITU: /mnt/c/Users/ankur/Desktop/SEM6/cc-lab8/PES2UG23CS077/raft_docker_lab$ docker stop node5
node5
ankur@JITU: /mnt/c/Users/ankur/Desktop/SEM6/cc-lab8/PES2UG23CS077/raft_docker_lab$ docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS          NAMES
9a1c1d61e5fe   raft_docker_lab-node1   "python node.py 1"      6 minutes ago   Up 6 minutes   node1
c6385eb8b544   raft_docker_lab-node3   "python node.py 3"      6 minutes ago   Up 6 minutes   node3
9bd2dcdb5543   raft_docker_lab-node2   "python node.py 2"      6 minutes ago   Up 6 minutes   node2
ankur@JITU: /mnt/c/Users/ankur/Desktop/SEM6/cc-lab8/PES2UG23CS077/raft_docker_lab$
```

SS15:

```
ankur@JITU: /mnt/c/Users/ankur/Desktop/SEM6/cc-lab8/PES2UG23CS077/raft_docker_lab$ docker stop node3
node3
ankur@JITU: /mnt/c/Users/ankur/Desktop/SEM6/cc-lab8/PES2UG23CS077/raft_docker_lab$ docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS          NAMES
9a1c1d61ef5e   raft_docker_lab-node1  "python node.py 1"      10 minutes ago  Up 10 minutes  0.0.0.0:2380->2380  node1
9bd2dcd85543   raft_docker_lab-node2  "python node.py 2"      10 minutes ago  Up 10 minutes  0.0.0.0:2380->2380  node2
ankur@JITU: /mnt/c/Users/ankur/Desktop/SEM6/cc-lab8/PES2UG23CS077/raft_docker_lab$
```

SS16:

```
[LEADER Node 1] I am the leader
[Node 1] Leader changed: None
[Node 1] Waiting for leader election...
[Node 1] Waiting for leader election...
[Node 1] Waiting for leader election...
[Node 1] Waiting for leader election...
[Node 1] Waiting for leader election...
[Node 1] Waiting for leader election...
[Node 1] Waiting for leader election...
```

SS17:

```
[FOLLOWER Node 2] Current leader: 172.28.0.2:8001
[Node 2] Leader changed: None
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
[Node 2] Waiting for leader election...
```

Q&A:

Leader election did not occur because Raft requires a majority (quorum) to elect a leader. In a 5-node cluster, at least 3 nodes must be active. With only 2 nodes running, consensus cannot be reached, so no leader is elected and the system remains unavailable.